How many benchmarks must be run? Selecting benchmarks and inferring the rest of a model's scores from a public score matrix

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Abstract

Evaluating a model on every benchmark of a large suite is costly, yet public leaderboards hold many "model $\times$ benchmark" score records. We ask: if only k benchmarks can be run for a new model, which k should they be, how should the remaining scores be inferred, and how large must k be? We study this on real, partially observed score matrices (MTEB embedding results at three data densities, Open LLM Leaderboard v2, HELM Lite, LiveBench) with a held-out-*model* protocol in which whole model families are held out and the choice of the k benchmarks never sees the held-out scores. We compare random subsets with greedy designs derived from a latent-factor model (Bayesian I-optimal design) and from an empirical Gaussian covariance (greedy variance reduction), and with k-means, pivoted-QR and communality-ranked choices, using factor-analysis, Gaussian-conditional and kNN predictors, and report family-cluster-bootstrap intervals. We also derive the error of random-subset prediction under a Gaussian factor model: an exact finite-k formula for least squares, a Jensen lower bound and a Marchenko–Pastur deterministic equivalent (1.4% median error against simulation) for the Bayes predictor, and we check it on real matrices. Findings: (i) the matrices are strongly low-rank-dominated (first PC 56–73% of variance) and, on MTEB, a random subset of 3–4 benchmarks closes half of the gap between the no-information baseline and the error reached with $k = 24$–48, while 8–16 are needed for 80%; (ii) choosing benchmarks matters most at small k (6–20% lower median error than random at $k \leq 4$) but the advantage of factor-model designs fades beyond $k \approx 8$–16, where the covariance-based greedy rule keeps a 6–19% edge and communality ranking becomes *worse* than random (+4 to +13%); (iii) the theory predicts the measured random-subset error curve of the factor predictor on MTEB to within about 20–35% but is pessimistic (measured 23–32% below implied) at $k = 1$ on the small LLM leaderboards; (iv) the benefit of designed subsets survives training on only $\approx$15 models but the pairwise-covariance predictor collapses when 80% of training cells are missing, while the EM factor model degrades gracefully. The LLM-leaderboard matrices have only 6–10 benchmarks, so our conclusions about k for generative LLMs are limited to $k \leq 6$.

1 Introduction

Public leaderboards accumulate a "model $\times$ benchmark" score matrix in which scores of different benchmarks are far from independent. An evaluator who must assess a new model but can afford to run only k of the N benchmarks faces three linked questions: *which* k benchmarks to run, *how* to infer the other $N - k$ scores from them and from the historical matrix, and *how large* k has to be.

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Item-level subsampling within one benchmark has been studied [Polo et al., 2024b, Vivek et al., 2024, Kipnis et al., 2024]; here we study the complementary benchmark-level problem, with the unit of selection being an entire benchmark (or task), and the unit of evaluation being a held-out *model*.

We state three hypotheses before running experiments. **H1** (low effective rank): because score matrices are dominated by a few latent factors, a small k suffices for most of the attainable accuracy. **H2** (design matters): subsets chosen by an experimental-design criterion beat random subsets, with an advantage that is largest when k is comparable to the effective rank and shrinks as k grows. **H3** (theory): under a Gaussian factor model the error of random-subset prediction follows a closed-form curve with a transition near $k \approx r$, and this curve describes real matrices. **H4** (data conditions): the conclusions hold when the history is small or sparse, but only for predictors that handle missing data properly.

Contributions. (1) A held-out-model, held-out-family evaluation protocol for benchmark-subset selection in which selection is blind to the new model's scores and hyper-parameters are tuned on training rows only. (2) A comparison of six selection rules and six predictors on six real matrices with cluster-bootstrap uncertainty. (3) Derivations of the random-subset error (Section 3.1) with synthetic verification and a check against real matrices, including where it fails. (4) A candid account of where the conclusions do and do not hold: they are supported on embedding benchmarks with up to 357 benchmarks, and only weakly testable on generative-LLM leaderboards that have 6–10 benchmarks.

2 Related work

Few-example evaluation. tinyBenchmarks selects or reweights evaluation *items* to estimate a model's accuracy on one benchmark [Polo et al., 2024b]; Anchor Points clusters items by model-confidence patterns [Vivek et al., 2024]; metabench distills six benchmarks into a sparse benchmark from the evaluations of thousands of models [Kipnis et al., 2024]. Our unit of selection is a whole benchmark and the target is the model's score on the *other* benchmarks. **Cross-benchmark structure.** BenchBench studies whether benchmarks agree and how methodological choices change that conclusion [Perlitz et al., 2024]; observational scaling laws explain scores on many benchmarks through a low-dimensional capability space estimated from public models [Ruan et al., 2024]; Sloth fits skill-based scaling laws to predict multi-benchmark performance across families [Polo et al., 2024a]. These works show that benchmark scores share a low-dimensional structure; we use the same structural assumption but ask the *design* question (which benchmarks to run) and quantify error as a function of k. **Statistical tools.** Factor analysis with missing data fit by EM [Rubin and Thayer, 1982, Tipping and Bishop, 1999], low-rank completion [Candès and Recht, 2012], Bayesian optimal design [Chaloner and Verdinelli, 1995] and greedy sensor placement in Gaussian models [Guestrin et al., 2005] are the building blocks; the contribution here is their combination and empirical assessment in the benchmark-selection setting, not a new estimator. The benchmark suites we use are MTEB [Muennighoff et al., 2022], HELM [Liang et al., 2022], the Open LLM Leaderboard and LiveBench.

3 Problem, methods and theory

Setting. A historical matrix $Y \in \mathbb{R}^{M \times N}$ (NaN = missing) of M models and N benchmarks. A new model has an unknown score vector $y \in \mathbb{R}^N$, of which the benchmarks C (those it has been evaluated on, in the data) are *candidates*. The evaluator chooses $S \subset C$, $|S| = k$ using only Y (selection is blind to y), observes y_S, and predicts $\hat{y}_T$ for $T = C \setminus S$. Columns are standardised with training-row statistics to z; error is $\mathrm{RMSE}_z = \sqrt{\frac{1}{|T|}\sum_{j \in T}(\hat{z}_j - z_j)^2}$ per held-out model, i.e. in units of the across-model standard deviation of each benchmark (also reported in score points).

Predictors. (a) *Train mean* ($\hat{z} = 0$; uses no scores; reference for "no information"). (b) *FA(r)*: a factor model $z = Vu + \varepsilon$, $u \sim \mathcal{N}(0, I_r)$, $\varepsilon_j \sim \mathcal{N}(0, \psi_j)$, fit by EM on the incomplete training matrix (80 EM iterations, SVD initialisation, loadings ridge 10^{-2}, $\psi_j \geq 0.02$; $r \in \{1, 2, 3, 4, 6, 8, 12, 16, 24\}$, $\alpha \in \{.02, .05, .1, .2, .4\}$), prediction $= V\,\mathbb{E}[u \mid z_S]$. (c) *Gauss-cov*: the Gaussian conditional mean $\Sigma_{\cdot S}(\Sigma_{SS} + 10^{-2}I)^{-1} z_S$ with Σ a pairwise-complete correlation matrix (min. 15 shared models, projected to the PSD cone) shrunk towards identity by α. (d) *kNN*: mean of the K=10 training

models closest on S. (e) PPCA (isotropic ψ) and a 1-factor model. r and α are chosen per fold on a validation split of the training rows (random subsets of sizes spanning the k grid).

Selection rules. *Random* (10 draws per model and k, averaged). *Greedy I-opt*: with fitted (V, ψ), the posterior covariance of u is $\Sigma_S = (I + V_S^\top \Psi_S^{-1} V_S)^{-1}$; greedily add the benchmark minimising $\mathrm{tr}(V^\top V\, \Sigma_S)$, the total posterior variance of the signal over all benchmarks (Bayesian A-optimal design; rank-one updates; $O(Nkr^2)$). Nested in k. We claim no approximation guarantee. *Greedy cov*: model-free greedy variance reduction on the shrunk correlation matrix, targets $= C$. *Top communality*: rank benchmarks by $\|v_j\|^2/(\|v_j\|^2 + \psi_j)$. *Pivoted QR* on $\Psi^{-1/2}V$ (deterministic leverage-type selection; falls back to communality for $k > r$). *k-means cover*: k clusters in loading space, one member each. Each rule is restricted to the benchmarks the model was actually evaluated on.

3.1 Theory: error of random subsets under a Gaussian factor model

Assume $z_j = v_j^\top u + e_j$, $u \sim \mathcal{N}(0, I_r)$, $e_j \sim \mathcal{N}(0, \sigma^2)$, loadings $v_j \overset{iid}{\sim} \mathcal{N}(0, I_r)$ known, and S a uniformly random k-subset. Let $W = V_S^\top V_S \sim \mathrm{Wishart}_r(k, I)$.

Proposition 1 (Exact error of least squares)**.** *With $\hat{u} = (V_S^\top V_S)^{-1} V_S^\top z_S$ and $k > r+1$, the expected squared error on a fresh benchmark is $\sigma^2\big(1 + \frac{r}{k-r-1}\big)$; for $k \le r+1$ it is infinite (and for $k \le r$ the estimator is not defined).*

Proposition 2 (Bayes predictor)**.** *With $\hat{u} = (\sigma^2 I + W)^{-1} V_S^\top z_S$ the expected error is $\sigma^2 + \mathbb{E}\ \mathrm{tr}\big[(I + W/\sigma^2)^{-1}\big]$, which (i) is at least $\sigma^2\big(1 + \frac{r}{k+\sigma^2}\big)$ by Jensen (matrix inversion is operator convex), (ii) is finite for every $k \ge 0$, (iii) satisfies, for $r, k \to \infty$ with $r/k \to c$, $\mathbb{E}\ \mathrm{tr}[(I + W/\sigma^2)^{-1}] \simeq r\lambda\, m(-\lambda)$ with $\lambda = \sigma^2/k$ and $m(-\lambda) = \frac{-(\lambda+1-c)+\sqrt{(\lambda+1-c)^2+4c\lambda}}{2c\lambda}$ the Stieltjes transform of the Marchenko–Pastur law [Marčenko and Pastur, 1967] at $-\lambda$. Limits: $\to \sigma^2 r/k$ for $k \gg r$, and $\to r - k$ for $k \ll r$ with $\sigma^2 \to 0$.*

Proofs are in Appendix A. The consequence for the evaluator is a *phase transition* at $k \approx r$: below it, unresolved latent directions keep (nearly) their prior variance, above it the excess shrinks like $\sigma^2 r/k$. For a *given* subset, the error under a fitted factor model is $\frac{1}{|T|}\sum_{j \in T}\big(\psi_j + v_j^\top \Sigma_S v_j\big)$ ("model-implied error"), which is exactly the quantity that the I-opt rule minimises (up to the choice of T). When loadings are estimated from M rows the error has an additional term, which we measure rather than derive (Appendix B).

4 Experimental protocol

Data. All from the pinned versions in `data.md`. **MTEB** (embeddings-benchmark/results at commit `6dabbcc`): one score per (model, task) = mean over subsets of `main_score` on the test (else validation/dev) split, averaged over a model's result directories; tasks with ≥ 100 models and models with ≥ 20 such tasks give *MTEB-wide* (291×357, 58% observed; the 234 models with ≥ 45 observed tasks are held out in turn, 70% observed). Two complete sub-matrices are extracted from it by greedily deleting the row or column with most missing cells (row scores multiplied by $\alpha = 1$ for *MTEB-dense* and $\alpha = 0.6$ for *MTEB-mid*; see `data_prep.py`): *MTEB-dense* (117×287) and *MTEB-mid* (139×65). **Open LLM Leaderboard v2** (4496 unflagged, de-duplicated models $\times$ 6 raw scores), **HELM Lite v1.13.0** (91×10), **LiveBench** judgments (mean per model and task; 7 tasks, 195 models, 318 missing cells; models with ≥ 6 tasks evaluated). The three MTEB matrices are nested views of the same data, not independent replications; the LLM boards are independent but tiny in N.

Splits. 10-fold cross-validation over *models*. *Family split* (primary): all models sharing a family key (organisation prefix for MTEB/OLB, first token of the model name for HELM/LiveBench) are held out together, folds balanced by size; *random split* for comparison. For every held-out model and every k in $\{1, 2, 3, 4, 5, 6, 8, 10, 12, 16, 20, 24, \dots\}$ up to $k_{\max}$ (24, 48, 32, 3, 6, 4 for the six matrices) all rules choose S from the model's observed benchmarks; the remaining observed ones are targets (at least 8 for MTEB, 3 for OLB/HELM, 2 for LiveBench; a held-out model must have

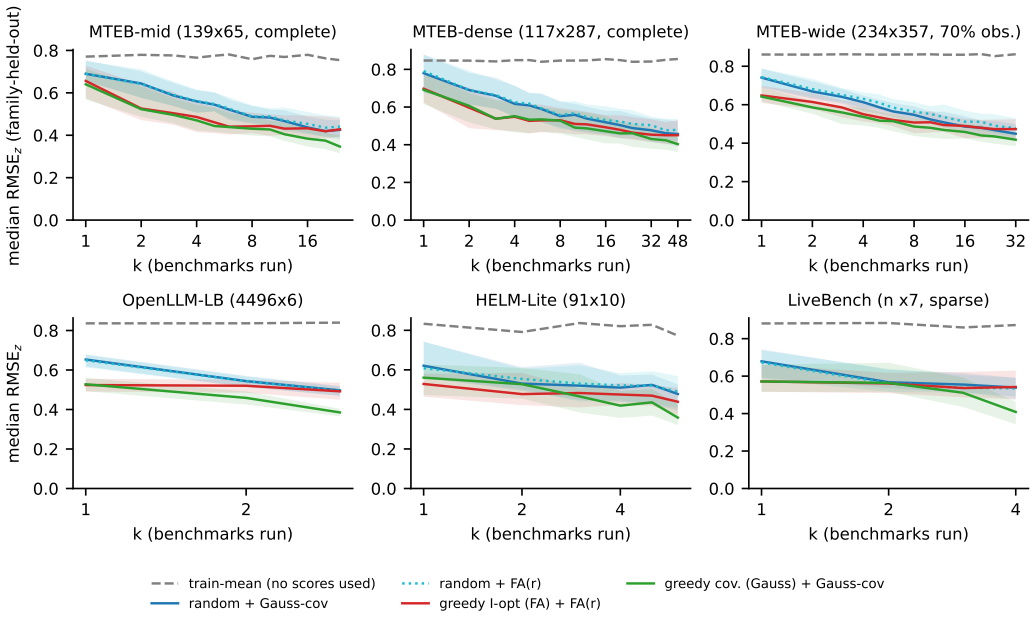


Figure 1: Median RMSE_z over held-out models (family split) vs. number of benchmarks run k; bands: 95% cluster-bootstrap CI. "Train mean" uses none of the new model's scores.

observed at least $k_{\max}$ plus that many benchmarks, which is why MTEB-wide is restricted to models with ≥ 45 tasks). Standardisation, EM, correlation estimation, and the selection of r, α use training rows only.

Statistics. Headline: median over held-out models of RMSE_z (a few new-family models are gross outliers, so means are also reported in the repository and quoted where they differ), with 95% percentile intervals from a cluster bootstrap over families (1000 resamples); paired comparisons use the same resamples. Wilcoxon signed-rank tests with Holm correction over all (matrix, split, rule, k) comparisons are descriptive only because models within a family are dependent. Fold-assignment seeds 0–2 are compared for MTEB-mid in Table 8. k_{50}/k_{80} denote the smallest k closing 50%/80% of the gap between the train-mean baseline and the error at $k_{\max}$ (so they depend on $k_{\max}$).

5 Results

5.1 A few benchmarks carry most of the signal (H1)

Table 1 shows strong dominance of the first principal component (56–73% of variance) and small participation ratios (1.8–3.0), although 90% of variance needs 9–23 components on MTEB. Figure 1 and Table 2 show that the median RMSE_z falls from 0.76–0.86 for the train-mean baseline to 0.43–0.55 at $k = 8$ and 0.35–0.49 at $k = 24$ on MTEB (family split; range over the three matrices and over random/designed subsets). Table 5: random subsets need $k_{50} = 3$–4 and $k_{80} = 8$–16 on MTEB; in raw units (Table 7) the median mean-absolute error on MTEB-mid drops from 9.6 score points (train mean) to 5.0 at $k = 8$ with random choice and to 4.2–4.4 with a designed subset. On OLB/HELM/LiveBench, $k_{\max} = 3, 6, 4$ already removes about 40% (random subsets) to 55% (greedy cov) of the baseline error (Table 6), but $k_{\max}$ is so small that k_{50} and k_{80} there are weakly informative.

5.2 Selection matters most at small k (H2)

Table 3 and Figure 2 give the paired change of median error relative to random subsets under the same Gauss-cov predictor. On the three MTEB matrices both greedy rules beat random, with intervals excluding zero in 41 of 42 (matrix, rule, $k \leq 8$) cases (the exception is I-opt at $k = 1$ on MTEB-mid, $[-17, +0.5]$%); the median gain is 6–20% at $k \leq 4$ and 4–12% at $k = 8$. Beyond $k \approx 8$–16 the I-opt

Table 1: Matrix summaries (z-scored correlation spectrum; pairwise-complete correlations where cells are missing). MTEB-wide: 291 models, of which the 234 with ≥ 45 observed tasks serve as held-out models. r_{80}, r_{90}: components for 80%/90% of variance.

matrix	models	benchm.	observed	PC1	r_{80}	r_{90}	part. ratio
MTEB-mid	139	65	100%	0.62	4	9	2.5
MTEB-dense	117	287	100%	0.56	5	15	2.9
MTEB-wide	291	357	58%	0.56	8	23	3.0
OpenLLM-LB	4496	6	100%	0.73	2	3	1.8
HELM-Lite	91	10	100%	0.63	2	4	2.3
LiveBench	195	7	77%	0.73	2	4	1.8

Table 2: MTEB, family split: median RMSE_z [95% family-cluster CI] over held-out models.

k	1	2	4	8	16	24
MTEB-mid (139x65, complete)						
train mean	0.771 [0.657,0.960]	0.779 [0.653,0.958]	0.766 [0.657,0.951]	0.759 [0.658,0.952]	0.779 [0.655,0.959]	0.754 [0.655,0.953]
random + Gauss	0.690 [0.654,0.754]	0.644 [0.577,0.701]	0.560 [0.506,0.612]	0.487 [0.453,0.538]	0.437 [0.406,0.487]	0.425 [0.381,0.470]
random + FA	0.690 [0.656,0.750]	0.641 [0.576,0.711]	0.561 [0.506,0.616]	0.489 [0.459,0.542]	0.454 [0.409,0.510]	0.441 [0.392,0.488]
greedy I-opt + FA	0.656 [0.574,0.728]	0.526 [0.471,0.641]	0.486 [0.415,0.554]	0.442 [0.409,0.507]	0.433 [0.388,0.493]	0.428 [0.394,0.481]
greedy cov + Gauss	0.640 [0.568,0.713]	0.523 [0.487,0.631]	0.470 [0.422,0.544]	0.431 [0.398,0.484]	0.384 [0.347,0.439]	0.347 [0.315,0.399]
MTEB-dense (117x287, complete)						
train mean	0.847 [0.696,1.024]	0.846 [0.697,1.021]	0.849 [0.695,1.026]	0.847 [0.697,1.025]	0.854 [0.697,1.023]	0.840 [0.694,1.027]
random + Gauss	0.780 [0.679,0.879]	0.690 [0.613,0.781]	0.617 [0.550,0.713]	0.552 [0.493,0.629]	0.519 [0.452,0.580]	0.489 [0.433,0.561]
random + FA	0.791 [0.679,0.878]	0.690 [0.615,0.792]	0.626 [0.547,0.713]	0.558 [0.483,0.630]	0.530 [0.458,0.604]	0.510 [0.451,0.583]
greedy I-opt + FA	0.697 [0.624,0.773]	0.597 [0.488,0.703]	0.548 [0.472,0.627]	0.530 [0.464,0.595]	0.492 [0.446,0.556]	0.464 [0.427,0.547]
greedy cov + Gauss	0.691 [0.616,0.783]	0.607 [0.518,0.674]	0.552 [0.465,0.620]	0.528 [0.458,0.586]	0.471 [0.407,0.552]	0.461 [0.388,0.514]
MTEB-wide (234x357, 70% obs.)						
train mean	0.861 [0.786,0.993]	0.860 [0.785,0.993]	0.861 [0.784,0.994]	0.861 [0.784,0.988]	0.860 [0.786,0.993]	0.853 [0.780,0.991]
random + Gauss	0.741 [0.688,0.787]	0.669 [0.630,0.722]	0.613 [0.567,0.647]	0.547 [0.512,0.591]	0.486 [0.457,0.540]	0.468 [0.433,0.516]
random + FA	0.743 [0.692,0.789]	0.681 [0.636,0.734]	0.630 [0.585,0.665]	0.564 [0.527,0.613]	0.512 [0.480,0.569]	0.490 [0.450,0.550]
greedy I-opt + FA	0.648 [0.609,0.698]	0.614 [0.578,0.661]	0.551 [0.513,0.614]	0.507 [0.451,0.562]	0.489 [0.426,0.526]	0.473 [0.431,0.523]
greedy cov + Gauss	0.642 [0.618,0.689]	0.586 [0.553,0.620]	0.537 [0.492,0.584]	0.486 [0.438,0.528]	0.458 [0.424,0.501]	0.435 [0.397,0.477]

design, which is tailored to the rank-r factor posterior, loses most of its edge (MTEB-mid: -20% at $k = 2$ but -3% and -5% at $k = 16, 24$), whereas the model-free covariance greedy rule keeps -6 to -19%. Pure communality ranking is the worst rule: it is better than random only at small k on MTEB-mid (and not reliably elsewhere), and its point estimate is 4–13% *worse* than random at $k = 16$ and 10–12% worse at $k = 24$ on the three MTEB matrices; inspection (Appendix C) shows it picks near-duplicate tasks (six CQADupstack subforums on MTEB-dense), i.e. redundancy is the failure mode. Holm-corrected Wilcoxon tests find the greedy-cov advantage significant at all tested k on the three MTEB matrices and the I-opt advantage at 36 of 40 (matrix, k) settings there (descriptive; Appendix D). On the LLM boards, designed subsets at $k = 1$ reduce error by 10–19% (e.g. OLB: MMLU-PRO is picked first by both greedy rules), but intervals are wide for HELM and LiveBench and the advantage at larger k is mixed (LiveBench $k = 4$: pivoted QR $+13\%$, I-opt -2%).

Predictors. With random subsets, Gauss-cov, FA(r) and kNN are within about 0.01–0.03 of each other for $k \geq 4$ on MTEB (e.g. MTEB-mid, $k = 8$: 0.487, 0.489, 0.496), a one-factor model is clearly worse (0.591), (Appendix Fig. 5). The gap that matters is between methods that use the new model's scores and the train mean, and between selection rules, not between sensible predictors. At large k the FA predictor saturates: its error cannot fall below the fitted noise level $\bar{\psi}$ per benchmark, while the Gauss-cov predictor, which exploits correlated *residuals* between sibling tasks, goes below it (MTEB-mid, random split, greedy cov + Gauss-cov, $k = 24$: trimmed MSE 0.089 vs. $\bar{\psi} = 0.096$). This is why factor-based design plateaus and a covariance-based design does not.

5.3 Theory against data (H3)

Synthetic check. Under the stated model, simulation (Appendix Fig. 3) reproduces Proposition 1 (e.g. $r = 6, \sigma^2 = 0.25$: simulated 1.015 vs. formula 1.000 at $k = 9$; the 99th percentile of the OLS error is 6.6 there, 16 at $k = 8$) and shows that the Bayes predictor tames the heavy tail (99th percentile 3.4 vs. 6.6 at $k = 9$). The Marchenko–Pastur formula has 1.4% median relative error

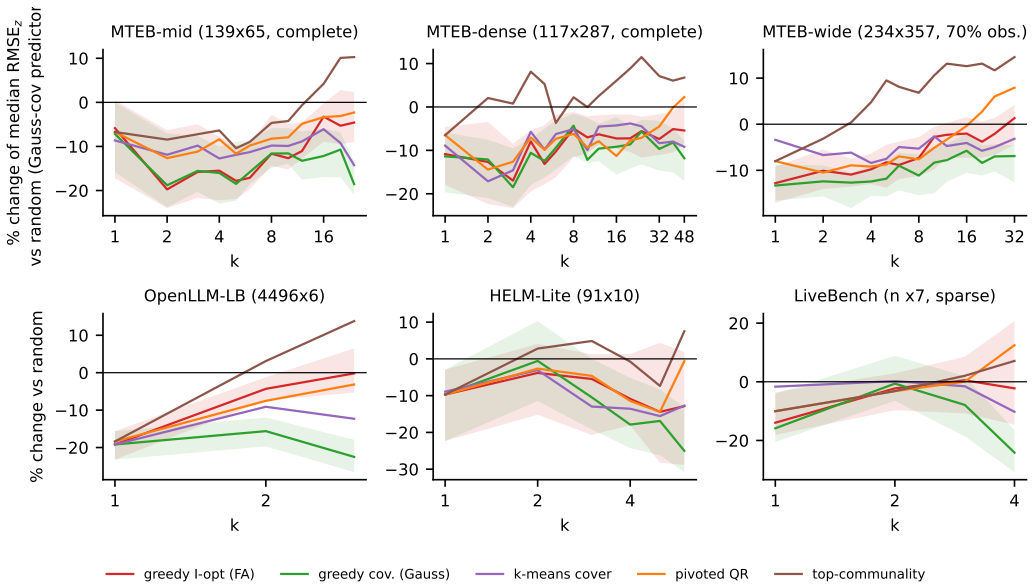


Figure 2: Change (%) of median RMSE_z relative to random subsets (same Gauss-cov predictor), family split; bands: 95% cluster-bootstrap CI for the two greedy rules.

Table 3: Selection rules vs. random (Gauss-cov predictor), family split: change of median RMSE_z [95% CI], percent.

matrix	selection	k=2	4	8	16	24
MTEB-mid	greedy cov	-19% [-24,-7]	-16% [-23,-10]	-12% [-15,-6]	-12% [-17,-6]	-19% [-21,-13]
MTEB-mid	greedy I-opt	-20% [-24,-6]	-15% [-22,-8]	-12% [-14,-3]	-3% [-9,-0]	-5% [-9,+2]
MTEB-mid	k-means cover	-12% [-17,-5]	-13% [-17,-6]	-10% [-13,-5]	-6% [-12,-3]	-14% [-20,-8]
MTEB-mid	pivoted QR	-13% [-17,-5]	-8% [-17,-2]	-8% [-11,-1]	-3% [-8,+0]	-2% [-8,+5]
MTEB-mid	top communality	-8% [-15,-3]	-6% [-14,-2]	-5% [-8,+1]	+4% [-1,+8]	+10% [+5,+15]
MTEB-dense	greedy cov	-12% [-21,-8]	-11% [-18,-7]	-4% [-11,-2]	-9% [-16,-2]	-6% [-13,-3]
MTEB-dense	greedy I-opt	-13% [-22,-3]	-8% [-19,-5]	-5% [-12,-1]	-7% [-10,-2]	-6% [-12,-2]
MTEB-dense	k-means cover	-17% [-25,-13]	-6% [-14,-0]	-5% [-11,-1]	-4% [-7,+1]	-4% [-9,-1]
MTEB-dense	pivoted QR	-14% [-23,-8]	-7% [-15,-5]	-6% [-12,-3]	-11% [-14,-4]	-7% [-11,-1]
MTEB-dense	top communality	+2% [-12,+15]	+8% [-6,+24]	+2% [-7,+6]	+6% [-1,+10]	+11% [+4,+18]
MTEB-wide	greedy cov	-12% [-16,-9]	-12% [-16,-9]	-11% [-15,-7]	-6% [-11,-2]	-7% [-10,-4]
MTEB-wide	greedy I-opt	-10% [-14,-5]	-10% [-13,-6]	-7% [-11,-3]	-2% [-7,+2]	-2% [-7,+1]
MTEB-wide	k-means cover	-7% [-11,-3]	-8% [-11,-3]	-5% [-8,-1]	-4% [-9,-2]	-5% [-8,-2]
MTEB-wide	pivoted QR	-10% [-15,-6]	-9% [-13,-5]	-8% [-11,-4]	-0% [-5,+4]	+6% [-0,+10]
MTEB-wide	top communality	-3% [-8,+5]	+5% [-0,+12]	+7% [+4,+16]	+13% [+5,+18]	+12% [+6,+20]

against simulation over 60 settings (max. 15% for $k \geq r$) versus 48% for the naive $\sigma^2 r/k$. Design value (T3): for $r = 6$, greedy I-opt reduces MSE relative to random by 42% (homogeneous loadings) to 90% (strongly heterogeneous benchmarks) at $k = r$, and the gain shrinks to 8–13% at $k = 6r$. *Real matrices.* Table 4 compares the measured (10%-upper-trimmed mean) MSE of random subsets with FA predictor with the model-implied error computed from the fitted factor model of each fold (no held-out scores used), random split. On MTEB the implied curve tracks the measured one: measured/implied ratios are 0.81–1.17 at $k = 4, 16$ (mean log-ratio over k: -0.10 MTEB-mid, -0.13 MTEB-wide, $+0.10$ MTEB-dense), with larger deviations at $k = 1$ (measured 12–20% below implied) and for MTEB-dense at $k = 24, 32$ (measured 26–36% above implied; we did not diagnose the cause). On OLB, HELM and LiveBench the measured error at $k = 1$ is 23–32% *below* the implied one (the model is pessimistic). For the Gauss-cov predictor (implied error = conditional variance of the shrunk correlation matrix) the measured error is on average 18–35% below the implied one across k and matrices (Appendix Fig. 4), as expected since shrinkage towards the identity inflates the implied variance. H3 is therefore supported *qualitatively* (shape, transition, ordering of rules) and *quantitatively* only to within tens of percent.

Table 4: Measured / model-implied MSE_z for random k-subsets with the FA predictor (random split; measured = 10%-trimmed mean over held-out models). $\bar{\psi}$: mean fitted noise variance.

matrix	k=1	k=4	k=16	$\bar{\psi}$
MTEB-mid	0.486 / 0.607	0.291 / 0.342	0.169 / 0.174	0.096
MTEB-dense	0.585 / 0.663	0.355 / 0.358	0.217 / 0.185	0.079
MTEB-wide	0.546 / 0.679	0.356 / 0.441	0.233 / 0.242	0.164
OpenLLM-LB	0.412 / 0.538	–	–	0.205
HELM-Lite	0.422 / 0.623	0.273 / 0.348	–	0.209
LiveBench	0.415 / 0.562	0.264 / 0.381	–	0.275

Table 5: How many benchmarks are needed (family split): smallest k closing 50% / 80% of the gap between the train-mean error and the error at $k_{\max}$ [95% bootstrap CI]. For OLB/HELM/LiveBench $k_{\max} \leq 6$, so these are weakly informative.

matrix	method	k_{50}	k_{80}	$k_{\max}$ tested
MTEB-mid	random+Gauss	3 [2,6]	8 [6,12]	24
MTEB-mid	greedy cov+Gauss	2 [1,4]	8 [4,16]	24
MTEB-mid	greedy I-opt+FA	2 [1,3]	4 [2,6]	24
MTEB-dense	random+Gauss	4 [2,6]	16 [8,24]	48
MTEB-dense	greedy cov+Gauss	2 [1,4]	10 [8,24]	48
MTEB-dense	greedy I-opt+FA	2 [1,3]	5 [2,16]	48
MTEB-wide	random+Gauss	3 [2,4]	10 [8,12]	32
MTEB-wide	greedy cov+Gauss	2 [1,2]	8 [5,10]	32
MTEB-wide	greedy I-opt+FA	1 [1,3]	4 [3,6]	32
OpenLLM-LB	random+Gauss	1 [1,2]	2 [2,2]	3
OpenLLM-LB	greedy cov+Gauss	1 [1,1]	2 [2,3]	3
OpenLLM-LB	greedy I-opt+FA	1 [1,1]	1 [1,1]	3
HELM-Lite	random+Gauss	1 [1,2]	2 [2,6]	6
HELM-Lite	greedy cov+Gauss	1 [1,3]	4 [3,6]	6
HELM-Lite	greedy I-opt+FA	1 [1,1]	2 [1,5]	6
LiveBench	random+Gauss	1 [1,2]	2 [2,2]	4
LiveBench	greedy cov+Gauss	1 [1,1]	4 [3,4]	4
LiveBench	greedy I-opt+FA	1 [1,1]	1 [1,2]	4

5.4 Data conditions (H4) and how large k must be

Table 9 (MTEB-mid, random split) varies the history. Keeping only 12% of the training rows ($M \approx 15$) leaves the median error with designed subsets at $k = 12$ at 0.61–0.64 versus 0.93 for the train mean on the same folds (Appendix Table 9), so the benefit of a few observed scores persists and the covariance-greedy design keeps a 10% edge over random at $k = 12$ (0.613 vs. 0.678), while the FA-based design stops improving beyond $k = 6$ (0.642 at $k = 6$, 0.643 at $k = 12$, 0.665 at $k = 24$). With 60% of training cells masked the designed-subset predictors degrade substantially (greedy cov + Gauss-cov at $k = 12$: 0.55 vs. 0.38 with full history) while FA-based ones degrade less (I-opt + FA: 0.50 vs. 0.39), and at 80% masked the Gauss-cov predictor collapses to the train mean: with few shared models per pair, the min.-overlap rule zeroes most correlations. This is an artefact of our pairwise-correlation estimator, not of the problem; the EM factor model still reaches $\mathrm{RMSE}_z \approx 0.62$ at $k = 12$ (vs. 0.84 for the mean). Fold-assignment seeds change medians by ≤ 0.02 for random and greedy-cov subsets and by up to 0.045 for greedy I-opt at $k = 4$ (Table 8), and the family split costs 0.02–0.05 relative to a random split, so family-level leakage is not what drives the MTEB results (but near-duplicate models from different organisations remain, see Section 6).

Answer to "how large must k be?" On MTEB: $k \approx 3$–4 for half of the achievable error reduction, $k \approx 8$–10 with covariance-greedy design (8–16 random) for 80% (the k_{80} values of different rules are not comparable because each is relative to that rule's own error at $k_{\max}$), and no tested $k \leq 48$ reaches a plateau with covariance-based design (median RMSE_z 0.40 at $k = 48$ on MTEB-dense, vs. 0.85 for the mean baseline). Consistent with Proposition 2, the relevant scale is $k \approx r_{\mathrm{eff}}$ (participation ratio 2.5–3, $r_{80} = 4$–8), with a long tail set by idiosyncratic benchmark variance.

6 Discussion and limitations

What the evidence supports. H1 is supported on MTEB: low effective rank means few benchmarks capture most of the recoverable signal. H2 is supported at small k and refuted as a general statement: the advantage of a factor-model design is concentrated near $k \approx r$ and disappears afterwards; communality ranking is harmful. H3 is supported in shape, not in constants. H4 is partly supported: designed subsets help with a small history, but conclusions depend on a predictor that handles missingness. **Limitations.** (1) The benchmarks are embedding tasks; the three generative-LLM matrices have only 6, 10 and 7 benchmarks, so $k \leq 6$ and claims about k for generative LLMs are weak, and families in HELM/LiveBench are defined by a crude name heuristic. (2) The dense MTEB matrices are selected by coverage, i.e. popular tasks and heavily evaluated models, which may inflate predictability; MTEB-wide (70% observed) is the least selected view. (3) A model's scores on the *target* benchmarks are those that were reported; selection bias in what authors report is not modelled. (4) Near-duplicate models from different organisations (fine-tunes, re-uploads) are not removed and may make the family split optimistic. (5) The median hides a heavy tail: the mean MSE_z of random+FA on MTEB-mid at $k = 1$ (family split) is 1.40 versus median 0.48, driven by new-family models with extreme scores (e.g. one model family scores up to 39 standard deviations, computed on the other models, above them on a language-specific task); no method here protects against such models. (6) The r grid is capped at 24 and the selected r often hits the cap on MTEB-mid/dense, so the FA fits are likely under-parametrised there. (7) Theory assumes Gaussian i.i.d. loadings, homoscedastic noise and a fresh benchmark; real loadings are clustered and heteroscedastic, the Marchenko–Pastur result is asymptotic, and our heuristic for the cost of estimating V from M rows (excess $\propto \sigma^2 r/M$) under-predicted the measured excess by 1.6–4.8$\times$ (the $1/M$ scaling itself held: log-log slopes $-0.87, -0.99, -1.06$ at $k = 5, 10, 20$). (8) The three MTEB matrices overlap and are not independent replications. (9) Scores are standardised per benchmark; rank-based or logit-scale models were not tried. (10) We do not compare with published methods for this specific task: baselines here are our own implementations of standard estimators and rules. **Reproducibility and honesty notes.** All numbers come from the scripts listed in the repository README; failed or unfinished items are listed there. Methods that were considered but not run: matrix-completion by nuclear-norm minimisation and gradient-boosted predictors.

7 Conclusion

For embedding benchmarks, running 3–4 well-chosen benchmarks recovers about half and 8–16 about 80% of the achievable error reduction for a new model's remaining scores, designed subsets beat random ones by 6–20% (median) at $k \leq 4$ but, for factor-model designs, little at large k, and a Marchenko–Pastur-based analysis explains the shape of the random-subset curve. On the small LLM leaderboards the same pattern appears for $k \leq$ 3–6 but cannot be tested further.

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A Proofs

Proposition 1. Write $z_S = V_S u + e_S$. Then $\hat{u} - u = W^{-1} V_S^\top e_S$ with $W = V_S^\top V_S$; conditional on V_S it is $\mathcal{N}(0, \sigma^2 W^{-1})$. For a fresh benchmark with $v \sim \mathcal{N}(0, I_r)$ independent of everything, $z - \hat{z} = e + v^\top(u - \hat{u})$ has mean-square $\sigma^2 + \mathbb{E}\|\hat{u} - u\|^2 = \sigma^2 + \sigma^2 \operatorname{tr} \mathbb{E} W^{-1}$. For $W \sim \mathrm{Wishart}_r(k, I_r)$, $\mathbb{E} W^{-1} = I_r/(k - r - 1)$ when $k > r + 1$ and the expectation diverges otherwise, giving the claim.

Proposition 2. With prior $u \sim \mathcal{N}(0, I)$, the posterior is $\mathcal{N}(\hat{u}, \Sigma_S)$ with $\Sigma_S = (I + W/\sigma^2)^{-1}$ and $\hat{u} = \Sigma_S V_S^\top z_S/\sigma^2 = (\sigma^2 I + W)^{-1} V_S^\top z_S$; since the model is correctly specified, the mean-square error of $\hat{z} = v^\top \hat{u}$ on a fresh benchmark, averaged over u, e and v, equals $\sigma^2 + \mathbb{E} \operatorname{tr} \Sigma_S$. (i) $A \mapsto A^{-1}$ is operator convex on positive definite matrices, so $\mathbb{E}(I + W/\sigma^2)^{-1} \succeq (I + \mathbb{E} W/\sigma^2)^{-1} = (1 + k/\sigma^2)^{-1} I$; taking the trace gives $r\sigma^2/(\sigma^2 + k)$. (ii) $\operatorname{tr} \Sigma_S \le r$. (iii) $(I + W/\sigma^2)^{-1} = \lambda(\lambda I + W/k)^{-1}$ with $\lambda = \sigma^2/k$; the empirical spectral distribution of W/k converges to the Marchenko–Pastur law with ratio $c = r/k$, so $\frac{1}{r} \operatorname{tr}(\lambda I + W/k)^{-1} \to m(-\lambda)$, which for the MP law is the positive root of $c\lambda\, m^2 + (\lambda + 1 - c)m - 1 = 0$, i.e. the stated expression. Limits follow by expanding $m(-\lambda)$ for $c \to 0$ (giving $r\lambda = \sigma^2 r/k$) and for $\lambda \to 0, c > 1$ (giving $m(-\lambda) \approx (1 - 1/c)/\lambda$, i.e. excess $r - k$). The statement is asymptotic; Appendix B reports finite-(r, k) accuracy. □

B Synthetic verification, selected benchmarks and additional results

Figure 3: (a) OLS (Prop. 1) and Bayes errors with 99th percentiles, 4000/4 trials per point; (b) MP formula vs. simulation, 3000 trials per point; (c) greedy I-opt vs. random under heterogeneous loadings/noise (60 instances per heterogeneity level, 200 simulated models each, loadings known); (d) excess MSE from estimating loadings by EM from M simulated models ($N = 100$, $r = 5$, $\sigma^2 = 0.3$, 8 repetitions, 0 or 50% missing), against the effective number of rows $M(1 - p)$. All data in this figure are *simulated* from the stated model.

C Benchmarks chosen by each rule (in-sample illustration)

Fitted on all rows (not used in any accuracy claim). OLB: greedy I-opt MMLU-PRO, GPQA, MATH Lvl 5; greedy cov MMLU-PRO, IFEval, MUSR; communality BBH, MMLU-PRO, GPQA. HELM Lite ($k = 4$): greedy cov MMLU, NarrativeQA, NaturalQuestions (closed-book), WMT 2014; communality MMLU, OpenbookQA, MedQA, NaturalQuestions (open-book). LiveBench: greedy cov coding_completion, paraphrase, connections, typos. MTEB-mid ($k = 6$, $r = 8$): I-opt WikipediaRetrievalMultilingual, NFCorpus, MedrxivClusteringS2S.v2, STS15, AmazonReviewsClassification, MassiveScenarioClassification; communality MassiveScenarioClassification,

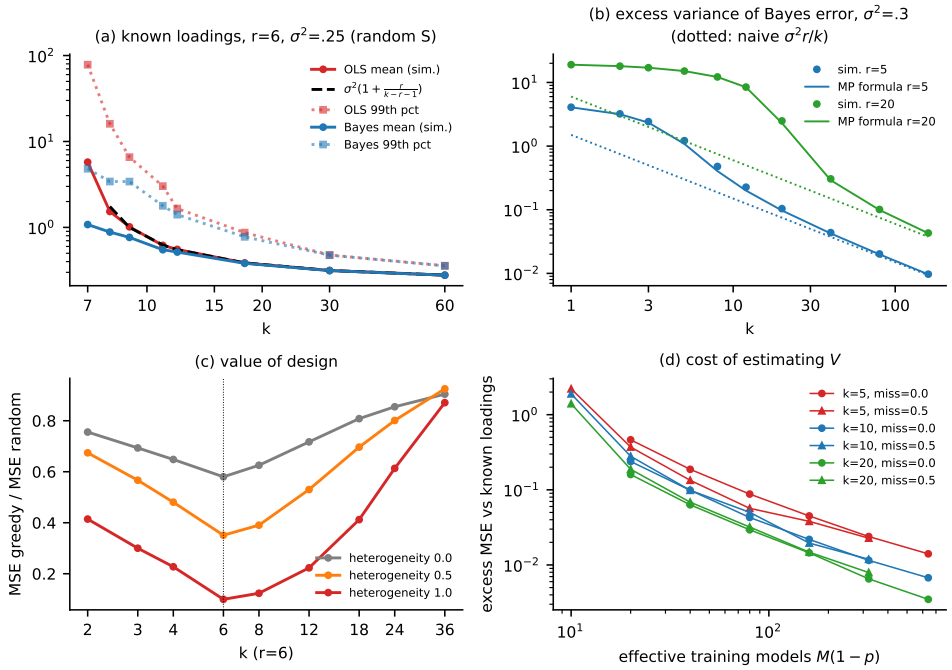


Figure 3: Synthetic checks of the theory (simulated data).

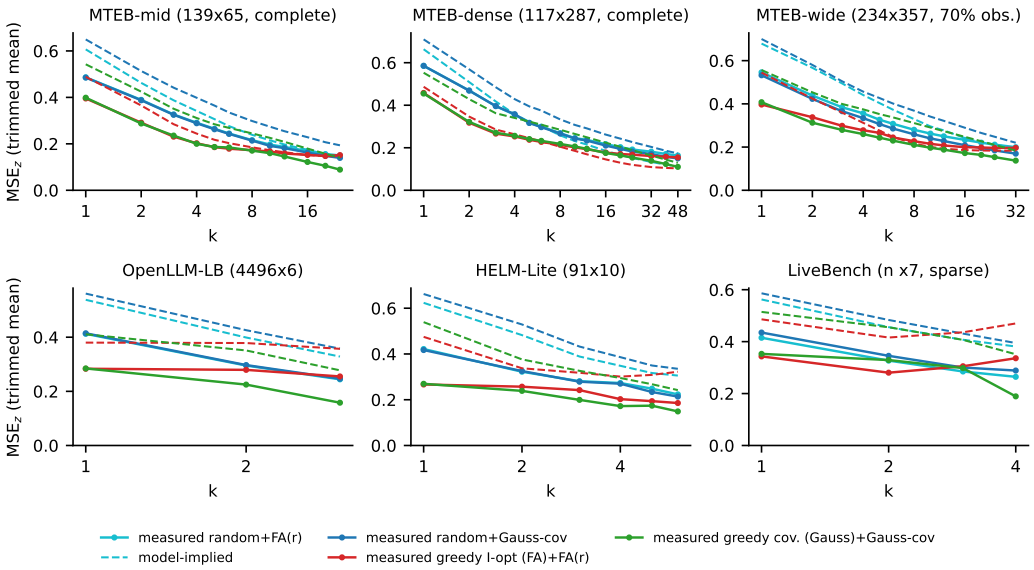


Figure 4: Measured (solid) vs. model-implied (dashed) MSE_z on real matrices, random split.

BiorxivClusteringP2P.v2, MedrxivClusteringP2P.v2, MedrxivClusteringS2S.v2, STS14, CQADupstackGamingRetrieval. MTEB-dense: communality picks six CQADupstack subforum retrieval tasks. Full output: `results/summary_selected.json`.

D Paired tests

`results/summary_paired_tests.csv` lists, for each matrix, split, rule ($\in$\{greedy I-opt, greedy cov\}) and k, the number of held-out models, mean/median paired difference of RMSE_z to random

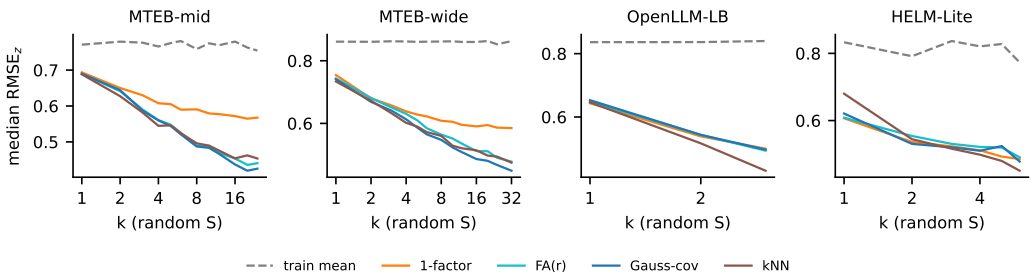


Figure 5: Predictors with random subsets (family split, median RMSE_z).

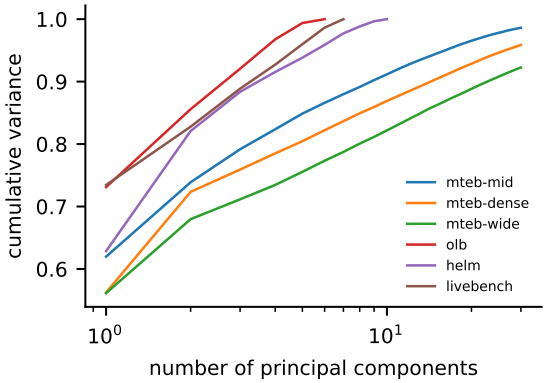


Figure 6: Cumulative variance of the correlation spectrum of each matrix.

(Gauss-cov) and the Holm-adjusted Wilcoxon p-value over all comparisons. Because models of one family are dependent these p-values are optimistic; the cluster-bootstrap intervals in Table 3 are the primary uncertainty statement.

E Compute, software, data licences

CPU only (40 cores available; all experiments $\approx$ 40 min wall-clock; synthetic checks $\approx$ 3 min). Python 3.11, numpy 2.4, scipy 1.17, pandas 3.0, scikit-learn 1.9, pyarrow 26, joblib 1.6. MTEB results: CC0-1.0. Open LLM Leaderboard and LiveBench dataset cards declare no licence; HELM Lite data follow the licences of the underlying scenarios (not verified item-by-item); only aggregate score tables are used.

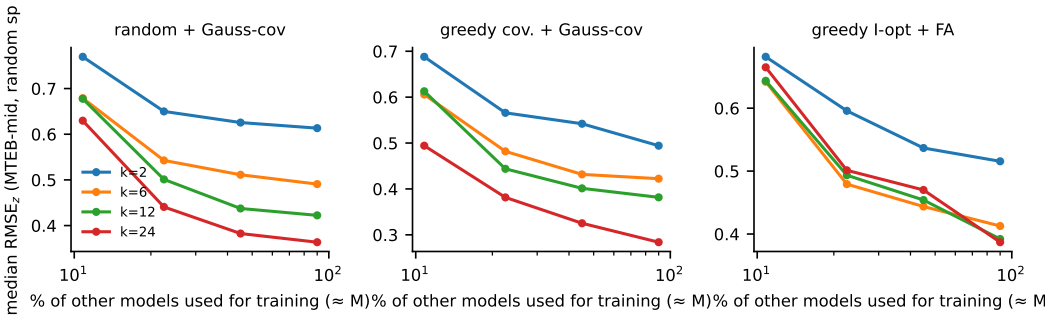


Figure 7: MTEB-mid (random split): median RMSE_z vs. fraction of training rows kept, for selected k.

Table 6: Generative-LLM leaderboards (family split), median RMSE_z [95% CI].

k	1	2	3
OpenLLM-LB (4496x6)			
train mean	0.836 [0.746,0.963]	0.837 [0.744,0.965]	0.840 [0.744,0.954]
random + Gauss	0.653 [0.617,0.679]	0.544 [0.515,0.570]	0.497 [0.470,0.519]
random + FA	0.649 [0.613,0.677]	0.543 [0.514,0.567]	0.494 [0.469,0.516]
greedy I-opt + FA	0.524 [0.491,0.556]	0.520 [0.485,0.551]	0.492 [0.449,0.535]
greedy cov + Gauss	0.528 [0.493,0.560]	0.459 [0.425,0.492]	0.385 [0.366,0.404]
HELM-Lite (91x10)			
train mean	0.833 [0.667,0.889]	0.791 [0.664,0.892]	0.837 [0.653,0.912]
random + Gauss	0.621 [0.560,0.742]	0.531 [0.494,0.614]	0.520 [0.448,0.636]
random + FA	0.609 [0.565,0.743]	0.554 [0.500,0.605]	0.531 [0.439,0.623]
greedy I-opt + FA	0.529 [0.465,0.580]	0.477 [0.422,0.585]	0.483 [0.410,0.506]
greedy cov + Gauss	0.561 [0.473,0.627]	0.528 [0.455,0.610]	0.466 [0.384,0.578]
LiveBench (n x7, sparse)			
train mean	0.881 [0.792,0.996]	0.884 [0.787,0.996]	0.860 [0.797,1.018]
random + Gauss	0.679 [0.592,0.743]	0.567 [0.536,0.634]	0.555 [0.514,0.617]
random + FA	0.672 [0.585,0.738]	0.561 [0.516,0.639]	0.538 [0.511,0.607]
greedy I-opt + FA	0.572 [0.517,0.661]	0.561 [0.509,0.617]	0.536 [0.487,0.623]
greedy cov + Gauss	0.571 [0.515,0.661]	0.564 [0.516,0.671]	0.511 [0.436,0.592]

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Table 7: Median mean-absolute error in benchmark score points (family split).

matrix	k	train mean	random+Gauss	greedy cov+Gauss	greedy I-opt+FA
MTEB-mid	4	9.43	5.84	4.61	4.47
MTEB-mid	8	9.58	4.95	4.42	4.15
MTEB-mid	24	9.36	4.04	3.61	4.08
MTEB-dense	4	8.39	5.45	4.37	4.24
MTEB-dense	8	8.39	4.53	4.01	3.90
MTEB-dense	24	8.34	3.76	3.52	3.57
MTEB-wide	4	10.79	6.41	5.24	5.63
MTEB-wide	8	10.81	5.56	4.82	4.98
MTEB-wide	24	10.79	4.59	4.30	4.51

Table 8: Stability across fold-assignment seeds and split type (MTEB-mid, median RMSE_z).

	random+Gauss		greedy cov+Gauss		greedy I-opt+FA	
	k=4	k=12	k=4	k=12	k=4	k=12
family split, seed 0	0.560	0.467	0.470	0.405	0.486	0.431
family split, seed 1	0.549	0.458	0.476	0.406	0.500	0.433
family split, seed 2	0.558	0.448	0.485	0.399	0.457	0.435
random split, seed 0	0.536	0.422	0.433	0.382	0.440	0.392

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Table 9: Data-condition sweep, MTEB-mid, random split, median RMSE_z (train-mean baseline ≈ 0.74 at full history; the baseline changes with the retained rows because standardisation is refit).

	random+Gauss		greedy cov+Gauss		greedy I-opt+FA	
	k=6	k=12	k=6	k=12	k=6	k=12
train rows kept: 12% of rows ($M \approx$15)	0.679	0.678	0.606	0.613	0.642	0.643
train rows kept: 25% ($M \approx$31)	0.543	0.501	0.482	0.444	0.479	0.493
train rows kept: 50% ($M \approx$62)	0.511	0.437	0.432	0.401	0.444	0.454
train rows kept: 100% ($M \approx$125)	0.491	0.422	0.422	0.382	0.413	0.392
train cells masked: 30%	0.530	0.463	0.490	0.442	0.433	0.421
train cells masked: 60%	0.617	0.581	0.576	0.551	0.490	0.503
train cells masked: 80%	0.827	0.835	0.845	0.856	0.609	0.621

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